ModelIt!

Student Activity Pack

Earth Systems Collide: Volcanoes, Climate, and Life

How a Volcanic Eruption Can Change the Whole Planet

Grade 05L2 | NGSS 5-ESS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What gases come out of a volcano when it erupts?

2 Do you think a volcano in one country can affect the weather in another country? Why or why not?

3 What are Earth's four major systems (spheres)? Name as many as you can.

4 What do you think 'negative feedback' means in science?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Volcanic Activity | Atmospheric CO2 | Global Temperature | Ocean Absorption | Vegetation Growth

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Earth

Low volcanic activity

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Massive Eruption

Volcanic Activity at 90%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Self-Regulation

Volcanic Activity at 60%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Volcanic Ash — Ash particles that block sunlight, temporarily cooling the planet

Ice Cap Melting — Glaciers that melt as global temperature rises

Human CO2 Emissions — Carbon dioxide added by humans burning fossil fuels

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How can a volcano on one continent change the climate on another? Use evidence from your simulation to support your answers.

1. **How does Earth regulate its own temperature naturally?**

2. **What surprised you about how vegetation responds to increased CO_2?**

3. **How is the volcanic CO_2 cycle similar to or different from human CO_2 emissions?**

4. **If Earth's self-regulation systems can handle volcanoes, can they handle human emissions too?**

5. **Why is it important to study how Earth's systems interact?**

8 STEM Challenge: Design an Early Warning Climate System

ModelIt!

YOUR ENGINEERING MISSION

Design a monitoring system that could detect and predict climate changes after a major volcanic eruption.

THE SCENARIO

A supervolcano like Yellowstone could erupt in the future. Your team needs to design a global monitoring system that tracks CO2, temperature, ocean chemistry, and vegetation changes to predict the climate impact.

Engineering Design Thinking

1. WHAT SENSORS WILL YOUR SYSTEM INCLUDE AND WHERE WILL THEY BE PLACED?
2. HOW WILL YOU MEASURE ATMOSPHERIC CO2, TEMPERATURE, OCEAN ABSORPTION, AND VEGETATION?
3. HOW WILL YOU DISPLAY THE DATA SO LEADERS CAN MAKE DECISIONS QUICKLY?
4. WHAT THRESHOLDS WILL TRIGGER WARNINGS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Volcanologists and Climate Modelers study how volcanic eruptions affect Earth's climate. They use computer models similar to what you built today to predict future climate changes. They earn

$70,000-$120,000/year.

9 Post-Assessment

ModelIt!

Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the Earth systems model, what happens when Volcanic Activity is set to 90%?

- **A.** Global Temperature drops because volcanic ash blocks sunlight
- **B.** Atmospheric CO_2 spikes, which causes Global Temperature to rise
- **C.** Ocean Absorption stops because volcanoes heat the water
- **D.** Vegetation Growth drops to zero because lava destroys all plants

2 What is a 'negative feedback loop' in the Earth systems model?

- **A.** A loop where one change causes more of the same change in an endless spiral
- **B.** A response that counteracts a change, pushing the system back toward balance
- **C.** A negative outcome that harms the environment permanently
- **D.** A loop that only occurs during volcanic eruptions

3 How does Vegetation Growth create a negative feedback loop with Atmospheric CO_2?

- **A.** More vegetation releases more CO_2 through photosynthesis
- **B.** When CO_2 rises, plants grow faster and absorb more CO_2 through photosynthesis, pulling CO_2 back down
- **C.** Vegetation blocks volcanic eruptions from releasing CO_2
- **D.** Plants die when CO_2 increases, which removes plant matter from the system

4 In the Self-Regulation scenario (Volcanic Activity at 60%), what does the model demonstrate about Earth's climate?

- **A.** Earth's temperature spirals out of control with any volcanic activity
- **B.** Vegetation and ocean absorption fully cancel out all volcanic CO_2
- **C.** The system reaches a new, higher equilibrium because feedback loops partially counteract the CO_2 input
- **D.** Earth's temperature is completely unaffected by volcanic eruptions

5 How can a volcanic eruption in Asia affect the climate in North America?

- **A.** It cannot; volcanic effects stay in the local area only
- **B.** Volcanic gases and CO_2 spread through the atmosphere globally, affecting temperature and weather patterns worldwide
- **C.** The eruption sends lava across the ocean to other continents
- **D.** Ash falls only on nearby countries and does not travel far